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SymAD-ECNN: Geometry Aware Equivariant Convolutional Neural Network based enhancements for Anomaly Detection in Biomedical Imaging Data

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LIST OF ABBREVIATIONS

AD - Anomaly detection

ML - Machine learning

DL - Deep Learning

CNNs - Convolutional Neural Networks

RNNs - Recurrent Neural Networks

MRI - Magnetic Resonance Imaging

CT - Computed Tomography

ECNNs - Equivariant Convolutional Neural Networks

SHAP - SHapley Additive exPlanations

AEs - Autoencoders

VAEs - Variational Autoencoders

GANs - Generative Adversarial Networks

RMSE - Root Mean Squared Error

MSE - Mean Squared Error

DSC - Dice Similarity Coefficient

AUROC - Area Under the Receiver Operating Characteristic Curve

AUPRC - Area Under the Precision-Recall Curve

CHAPTER 01: INTRODUCTION

1.1 Chapter overview

This chapter provides a comprehensive foundation for the research by presenting the background and domain of deep learning for anomaly detection, with a focus on biomedical imaging. It defines the core problem and articulates its real-world relevance and challenges. Motivated by the limitations of current deep learning methods, this chapter reviews existing work, identifying significant gaps that hinder effective anomaly detection in complex, multimodal, and spatial data. The contribution of this research to advancing the scientific body of knowledge is outlined, emphasising the integration of symmetry-aware geometric deep learning within anomaly detection frameworks. Key research challenges, questions, and the overarching aim of the project are clearly stated, guiding the study. This overview sets the stage for detailed exploration in subsequent chapters, ensuring clarity and focus throughout the thesis.

1.2 Problem background

1.2.1 The growing significance of anomaly detection

An anomaly is defined as an observation that significantly deviates from the expected behaviour in a system, often appearing unusual, inconsistent, or unexpected. Outliers, despite constituting a small fraction of a dataset, are crucial as they carry important information and can reveal critical insights during analysis (Huang et al., 2025). Anomaly detection (AD) is the process of identifying these anomalous observations using various methods and algorithms, which help decision-makers better understand data patterns and behaviours. The rapid expansion of data from diverse sources has made anomaly detection increasingly essential. AD plays an increasingly important role across various communities, including data mining, machine learning, computer vision, and statistics, due to growing demand in broad domains such as risk management, compliance, security, financial surveillance, health and medical risk, and AI safety (A.B. Nassif et al., 2021). Traditionally, statistical methods and classical machine learning (ML) algorithms such as clustering, support vector machines, and isolation forests have been used to identify unusual patterns or outliers in data (L. Ruff et al., 2021).

1.2.2 The evolution of Deep Learning

Deep Learning (DL) has revolutionised AD, with advancements that have significantly enhanced anomaly detection methods, making them more powerful and adaptable, demonstrating significantly better performance than conventional anomaly detection on addressing challenging detection problems in a variety of real-world applications (Pang et al., 2020), especially for handling high-dimensional and unstructured data. Deep neural networks (DNNs) can autonomously learn complex patterns and hierarchical representations from raw data, which removes the need for labour-intensive feature engineering. This is particularly advantageous for detecting subtle and multifaceted anomalies that might be missed by traditional methods. DL models excel at processing various data types, including images, videos, and text, by capturing non-linear relationships and interactions (Huang et al., 2025), thus offering a more flexible and robust framework for AD. This has led to a significant shift from purely traditional AD techniques to deep learning methodologies with architectures like Autoencoders, Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) showing promising results (Landauer M. et al., 2023).

1.2.3 Anomaly detection in the biomedical domain

In healthcare, even a single undetected anomaly in medical imaging may represent the difference between early intervention and late diagnosis. The high stakes of these anomalies underline why robust detection methods are not just desirable, but essential. The often time-consuming, subjective, and variable nature of manual medical diagnosis makes computer-assisted interventions highly beneficial for increasing quality and decreasing costs, especially in areas with limited access to healthcare services. Despite these advancements, challenges persist which remain active research areas to improve clinical adoption and diagnostic accuracy.

1.2.3.1 Medical imaging and brain tumours

Anomaly detection has become a cornerstone in biomedical applications, particularly in the analysis of complex medical imaging and physiological data, where identifying rare and subtle abnormalities is critical for early diagnosis and treatment (Huang et al., 2025). Traditional machine learning methods, such as clustering and support vector machines, provided early frameworks for anomaly identification but struggled to capture complex patterns inherent to high-dimensional, multimodal biomedical data. The rise of deep learning has dramatically enhanced anomaly detection capabilities by leveraging powerful

architectures like convolutional neural networks (CNNs), recurrent neural networks (RNNs), autoencoders, and generative adversarial networks (GANs) that automatically learn hierarchical features directly from the data. For example, recent deep learning models applied to neurological MRI images have achieved a high rate of classification accuracy, indicating robust anomaly characterisation in diseases such as epilepsy and Parkinson's (R. Ranjbarzadeh, 2021). Brain tumours present some of the most demanding challenges in medical diagnostics due to their complex morphology, heterogeneous presentation, and critical location within the brain. Medical imaging, especially MRI and CT scan modalities, serves as the primary tool for detection and classification, traditionally relying on radiologist expertise and conventional image processing techniques (M. Aggarwal, 2023). While these classical methods facilitate rapid identification of gross anatomical anomalies, they often lack sensitivity to subtle tumour boundaries and variations in tumour type or growth patterns, leading to variable accuracy and reproducibility. Deep learning-based automated methods have increasingly taken center stage, showing improved performance by learning pixel-wise distinctions and structural patterns directly from imaging data (S.Poonkodi, 2023).

1.3 Problem definition

In clinical practice, a missed tumour in a scan may mean the difference between early intervention and terminal progression. Accurate anomaly detection in biomedical imaging, particularly brain tumour detection, remains a highly challenging problem due to the inherent complexity, heterogeneity, and variability of the data involved. Despite the advances in deep learning techniques, existing models often fall short in fully capturing the spatial and structural intricacies of brain anatomy. Traditional convolutional neural networks (CNNs) and other architectures typically lack geometric awareness, rendering them sensitive to transformations such as rotations, translations, and reflections that are common in clinical imaging data. This sensitivity undermines their robustness and generalisation across diverse datasets and imaging protocols, limiting their practical clinical utility. Moreover, interpretability remains a critical challenge. Clinicians require transparent and explainable models to trust and adopt these systems in real-world diagnostic workflows. This research aims to tackle these critical limitations by developing a novel anomaly detection framework that explicitly incorporates geometric priors to improve robustness and generalizability in brain tumour detection. The proposed method will utilise symmetry-aware learning principles to handle spatial transformations naturally present in biomedical imaging data. Furthermore, emphasis will also be placed on interpretability and explainability to promote clinical trust.

By addressing these interconnected challenges, the project seeks to advance the state-of-the-art in automated brain tumour detection while establishing a foundation for generalisation to other biomedical domains.

1.3.1 Problem statement

Anomaly detection in medical imaging remains a critical unsolved challenge, as current deep learning models rely heavily on extensive data augmentation, lack true spatial equivariance, require large, annotated datasets, and produce high rates of false positives with limited interpretability, leading to delayed or missed diagnoses that put patient survival and clinical decision-making at risk.

1.4 Research motivation

This study is driven by the urgency for more accurate and effective anomaly detection in medical imaging, as missed or incorrectly identified abnormalities, including cancers or lesions, can have a direct impact on patients' lives. Although deep learning has enhanced diagnostic workflows, current approaches compromise in biomedical contexts since they rely extensively on annotated datasets and synthetic augmentation. The author, drawing on a strong background and deep interest in mathematics, geometric and computation principles, was motivated to address these limitations through equivariant convolutional neural networks (ECNNs), which naturally encode invariance and reduce reliance on data augmentation and further, to create a generalizable framework that can be applied to other scientific domains in addition to the promise of increased accuracy and robustness in identifying medical anomalies, especially in intricate areas like brain tumour analysis.

1.5 Existing work

Citation	Summary	Contribution	Limitation
Pinaya W. et al. (2022)	The paper introduces a transformer-based unsupervised method for 3D brain anomaly detection and segmentation using VQ-VAE, achieving superior performance	Contributions are a novel combination of VQ-VAE and transformers for anomaly detection, utilising spatial resampling masks for improved	Limitations include potential missed detections if the VQ-VAE encoder "heals" anomalies, challenges in generalising 3D models, dependence

	on synthetic and real MRI datasets.	segmentation, and demonstrating state-of-the-art results with computational efficiency.	on large healthy datasets, and reconstruction quality affecting segmentation accuracy.
Han C. et al. (2020)	This paper introduces MADGAN, a novel unsupervised method using GAN-based reconstruction of multiple adjacent brain MRI slices to detect various brain diseases, including early and late-stage Alzheimer's disease and brain metastases, by leveraging large-scale healthy MRI data for anomaly detection.	The main contributions are the first use of multiple MRI slice reconstruction for anomaly detection, enabling unsupervised multi-stage and various disease diagnosis on multi-sequence MRI scans by predicting the next 3 slices from the previous 3 and discriminating abnormalities using average ℓ2 loss.	The main limitations are that the approach is less generalizable in both reconstruction and diagnosis, requiring further improvements in self-attention module placement and loss functions, as well as more healthy T1c scans for reliable detection of various diseases.
Abdusalomov A. et al. (2023)	This paper presents an enhanced YOLOv7 deep learning model, incorporating CBAM attention, SPPF+, and BiFPN modules, to achieve highly accurate detection and localisation of meningioma, glioma, and pituitary brain tumours in MRI images, outperforming	The main contribution is the development of a fine-tuned YOLOv7 model with integrated CBAM, SPPF+, and BiFPN modules, achieving 99.5% accuracy in brain tumour detection and surpassing previous methods in both accuracy and	A major limitation is the need for a large, well-annotated dataset, as bounding boxes may not capture precise tumour borders and YOLO may be sensitive to data shortages, requiring further validation on real clinical data.

	existing state-of-the-art methods.	localisation capability (Source).	
Ravinder, M et al. (2023)	This paper proposes a novel hybrid Graph Convolutional Neural Network combining CNN and GNN techniques to improve brain tumour classification accuracy from MRI images by modelling pixel relationships using graph convolution.	It introduces a graph-based convolutional approach that captures non-Euclidean pixel relationships in brain MRI data, achieving superior classification accuracy (95.01%) compared to conventional CNN models.	The model's high computational complexity and reliance on a fixed adjacency matrix limit its scalability and adaptability to varying image resolutions and tumour shapes.
Lei W. et al. (2024)	This paper presents a comprehensive benchmark for anomaly detection in medical images, evaluating thirty representative methods across seven datasets and five image modalities, and analyzes the effects of key methodological components to reveal unresolved challenges and guide future research.	The main contribution of this work is the establishment of a unified benchmark with seven curated medical datasets and thirty typical anomaly detection methods, providing a fair and comprehensive comparison, systematic analysis of key components, and identification of unresolved challenges and future directions in medical anomaly detection.	A key limitation of this study is that current strategies for latent space configuration and reconstruction error measurement in reconstruction-based methods remain sub-optimal, and the evaluation of pixel-level anomaly segmentation is limited by the scarcity of annotations and the narrow range of anomaly types considered.

Table 1. Existing work

1.6 Research gaps

- **Lack of spatial equivariance and high computational overhead in medical anomaly detection:** CNNs and traditional deep learning models in medical imaging struggle to maintain consistency across spatial transformations such as rotations and translations, especially in brain MRI, where tumours appear in diverse orientations (Huang et al., 2024). While data augmentation and transfer learning techniques have been applied to approximate invariances, which partially address this, it is an inefficient workaround. Meantime, most existing architectures increase training, inference time, computational costs and risk introducing artifacts (S. Poonkodi et al., 2023). Addressing this gap can lead to models that are robust and efficient, reducing training time and improving generalisation in clinical settings.

- **Over-reliance on large, annotated datasets:** Deep learning models for biomedical anomaly detection often demand extensive labelled datasets, which are costly and impractical to obtain in clinical settings. For rare conditions like brain tumours, annotations require expert radiologists, making data scarcity a bottleneck. Previous approaches attempted semi-supervised or synthetic data generation, but these remain prone to overfitting (Pang et al., 2023) and reduced generalizability. A model capable of learning effectively from smaller, multi-modal datasets without heavy annotation would address this gap (Ravinder, M et al., 2023).

- **High false positive rates due to class imbalance:** The rarity of anomalies compared to normal data points in most AD tasks results in class imbalance. This imbalance can bias the detection algorithms toward the majority class (normal data), potentially leading to a higher rate of false negatives (missed anomalies) or misclassifying normal data as anomalous due to insufficient representation of true anomaly characteristics. These can have serious consequences, such as unnecessary biopsies, delayed interventions, or patient anxiety. Conventional deep learning methods often flag benign variations as anomalies, leading to reduced trust in AI-assisted diagnostics. Current deep learning methods are especially vulnerable, as they tend to overfit to majority patterns and lack robustness against benign variations such as imaging artifacts or natural anatomical differences. Thus, reducing false positives while maintaining sensitivity to rare

anomalies remains a critical gap in medical anomaly detection research (Pang et al., 2023) (Baur, C. et al., 2020).

- **Lack of explainability for clinical trust and adoption:** Existing explainability efforts provide partial insights but lack integration with geometry-aware learning. As a result, the reasoning behind model predictions remains opaque, especially when using complex architectures like graph neural networks or equivariant models (Ahmedt-Aristizabal, D. et al., 2021). This lack of interpretability poses a major barrier to clinical adoption: physicians and radiologists need to understand not just *what* the model predicts, but also *why* it highlights certain structures as anomalous. Without explainble, geometry-informed outputs, AI systems risk being perceived as unreliable "black boxes." This gap highlights the urgent need for models that combine geometric awareness with built-in interpretability, enabling clinicians to validate model reasoning and build trust in AI-assisted diagnostics.

1.7 Contribution to the body of knowledge

1.7.1 Problem domain contribution

In the pressing clinical context of brain tumour detection, this research proposes solutions that align methodological innovation with urgent medical needs. Brain tumours are highly heterogeneous in shape, size, and spatial distribution, making conventional pipelines prone to false positives and reduced sensitivity. By **employing symmetry-aware architectures, this work naturally captures spatial variability** without excessive preprocessing, offering enhanced diagnostic reliability and resilience. This improvement directly addresses a major bottleneck in real-world clinical workflows, where variability in tumour presentation can lead to missed diagnoses or delayed treatment.

The research also mitigates the scarcity of annotated medical datasets by designing architectures that are **optimised for learning under data-limited conditions**. This addresses a fundamental barrier in clinical practice, where large-scale labelled data are seldom available due to privacy, cost, and time constraints. By ensuring reliable performance under such limitations, the work supports deployment across diverse healthcare systems, including those with limited resources.

Equally significant is the integration of explainability tools within the diagnostic process. By providing clinicians with visual and quantitative explanations of predictions, the **model enhances trust, facilitates clinical validation, and reduces the likelihood of misinterpretation**. This bridges the gap between automated anomaly detection and practical clinical decision-making, ensuring the system complements rather than replaces human expertise.

Collectively, these contributions strengthen the problem domain by delivering a clinically usable, trustworthy, and adaptable system for brain tumour diagnosis. The work not only addresses current bottlenecks in diagnostic reliability, data scarcity, and interpretability but also lays the foundation for extending these advances to other biomedical anomaly detection tasks in the future.

1.7.2 Research domain contribution

This research advances the evolving field of biomedical anomaly detection by directly addressing key limitations of existing deep learning approaches in brain tumour diagnosis and beyond. Standard convolutional neural networks (CNNs) and related models, while achieving promising performance in classification and segmentation, neglect intrinsic geometric properties such as spatial equivariance, thereby requiring extensive data augmentation and large annotated datasets to maintain robustness. By introducing **a novel symmetry-aware learning framework**, this work enables models to inherently respect rotations and translations present in MRI scans, eliminating the need for brute-force augmentation, hence reducing the risk of overfitting and a decreased number of false positives and while maintaining feasible computational costs.

In parallel, the research confronts the dependency on large-scale labelled datasets, which remain costly, time-intensive, and often fail to capture the full variability of clinical presentations. The proposed data-efficient architectures are designed to learn from limited, heterogeneous datasets, expanding accessibility of AI methods in resource-constrained healthcare environments. This not only strengthens the scalability of biomedical AI but also democratizes its adoption across diverse clinical contexts.

A further contribution lies in advancing interpretability within anomaly detection. By embedding explainability mechanisms such as SHAP and Grad-CAM, the research provides transparent decision pathways, addressing clinicians' growing demand for trustworthy AI.

Interpretability is not treated as an afterthought but as a core design principle, ensuring AI outputs align with medical reasoning.

1.8 Research challenges

- **Limited availability of annotated biomedical datasets:**

Medical imaging datasets with expert annotations are limited in both size and diversity. Annotating MRI volumes requires highly skilled radiologists and is time-consuming, making large-scale datasets difficult to produce. This scarcity of annotated datasets poses a major barrier to training deep learning models, which typically require large amounts of labelled examples. Without adequate data, models risk poor generalization and reduced reliability.

- **False positives and overfitting risks:**

In medical imaging, particularly with small or imbalanced datasets, models face the dual risk of generating false positives and overfitting. False positives can erode clinical trust, leading to unnecessary interventions, while overfitting results in models that perform well on training data but fail on unseen cases. Both issues are magnified by the complexity of 3D brain scans, where subtle variations in tumour shape or texture may confuse the model. This challenge is significant because a clinically deployable model must demonstrate both accuracy and reliability across diverse populations.

- **Explainability and clinical trust:**

Deep learning models are often criticized as "black boxes." In medical applications, explainability is not optional; it is essential for clinical adoption, as physicians must trust and understand the system's reasoning before acting on its outputs. Without explainability, even a highly accurate model could face resistance from healthcare practitioners and regulatory bodies. Embedding explainability mechanisms into equivariant models is, therefore, critical to bridge the gap between algorithmic output and medical reasoning.

- **Generalisation beyond brain scans:**

While this research focuses on brain tumour detection, a broader challenge is generalising models across different organs and medical imaging modalities. Most models are highly specialised for a single dataset or task, limiting their adaptability. From a research perspective, developing architectures that generalise well is crucial to advancing medical AI as a whole. This challenge is significant because healthcare systems demand versatile tools that extend beyond a narrow use case.

1.9 Research questions

RQ1: How can symmetry-aware deep learning architectures be designed to inherently model spatial equivariance in medical imaging data like brain MRI scans, and how does this improve anomaly detection robustness and generalisation compared to traditional CNN-based methods?

RQ2: How can deep learning models be optimised to maintain high diagnostic performance with limited annotated data in medical anomaly tumour detection, like brain tumours, ensuring accessibility and efficacy without relying on extensive data augmentation?

RQ3: In what ways can advanced explainability techniques be incorporated into anomaly detection frameworks to provide transparent, clinically meaningful insights that improve clinician trust and support decision-making?

1.10 Research aim

The aim of this research project is to design, develop, and evaluate a geometry-aware deep learning model based on equivariant convolutional neural networks (ECNNs) for reliable and explainable anomaly detection in medical imaging data, with a particular focus on brain tumour identification.

By utilising the inherent symmetries and structural characteristics of biomedical data, this project aims to increase accuracy, explainability and decrease false positives, in contrast to traditional deep learning techniques that mainly rely on large, annotated datasets and data augmentation, all with a viable computation cost. In addition to improving anomaly detection in biomedical data, the research aims to lay the foundation for a generalized model across domains, by including mathematical concepts of equivariance into the model design. This dual focus ensures that while the immediate clinical benefits are realized in brain tumour detection, the methodology retains adaptability for applications in other domains and broader scientific contexts. Ultimately, the project seeks to demonstrate that geometry-aware, data-efficient models can transform both domain-specific diagnostics and the wider field of artificial intelligence in healthcare. Beyond technical advancement, the research aspires to reduce barriers to clinical adoption by creating transparent and trustworthy AI systems, thus fostering stronger collaboration between computational research and medical practice.

1.11 Research objectives

Objectives	Research Objective	Description	LOs Mapped	RQ Mapped
Problem Identification	**RO1**: To investigate the clinical challenges in anomaly detection for brain MRI and other medical imaging modalities **RO2**: To analyze technical limitations of deep anomaly detection models in medical imaging.	Define the current challenges and unmet needs in medical imaging anomaly detection, focusing on issues like spatial equivariance, data scarcity, computational cost, and interpretability.	LO4, LO6	RQ2
Literature Review	**RO3**: To comprehensively go through research papers to learn about existing methodologies and approaches. **RO4**: To review and critically evaluate existing deep learning models for encoding spatial equivariance in medical imaging. **RO5**: To review data-efficient training techniques and interpretability methods in medical anomaly detection	Conduct a comprehensive review of existing methods and technologies in deep learning for medical imaging, including symmetry-aware models,data-efficient training, and explainability approaches.	LO4, LO5, LO6	RQ1, RQ2, RQ3
Requirement Elicitation	**RO6**: To identify the prerequisites on both clinical and technical aspects for deep anomaly	Gather and analyze the needs and constraints of stakeholders, including clinical and technical	LO2, LO3, LO6	RQ1, RQ2, RQ3

	detection including spatial invariance, adaptive multimodal fusion, data efficiency, and meaningful interpretability **RO7**: To determine necessary data for model training, and other software and hardware requirements.	requirements for developing a robust and interpretable anomaly detection system in medical imaging.		
System Design	**RO8**: To design symmetry-aware neural network architectures that inherently encode geometric transformations. **RO9**: To incorporate transparency and explainability mechanisms directly within the produced anomaly detection models	Develop architecture and fusion frameworks that embed symmetry awareness and interpretability based on elicited requirements.	LO1, LO2, LO5, LO6	RQ1, RQ2, RQ4
Implementation	**RO10**: To implement baseline models. **RO11:** To implement and train the model on benchmark datasets. **RO12**: To apply data-efficient training and domain adaptation techniques to optimize model performance	Build and train the models and systems, applying data-efficient techniques and modular coding practices to support adaptability and experimentation.	LO2, LO5, LO7	RQ1, RQ3, RQ4
Testing	**RO13**: To evaluate models on benchmark brain MR and other datasets	Evaluate model performance rigorously on benchmark datasets and real clinical scenarios, measuring	LO5, LO8	RQ1, RQ3

	RO14: To evaluate the framework against other baseline models	accuracy, robustness, interpretability, and data fusion effectiveness.		
Documentation	**RO15**: To document methodology, experiments, and results in a coherent thesis and publishable papers. **RO16**: To improvise thesis and papers with supervisor feedback.	Improvise reports and papers documenting the research process, methods, findings, challenges, and contributions with feedback from mentors and experts.	LO5, LO8,	RQ1, RQ2, RQ3, RQ4
Publication	**RO17**: To publish a review paper on related work. **RO18**: To publish a conference paper on related work.	Draft a review paper and conference paper on related work and publish	LO7, LO8, LO9	RQ1, RQ2, RQ3, RQ4

Table 2. Research objectives

1.12 Project scope

The scope outlines the components and activities included within the study and specifies the areas that are excluded to maintain focus and feasibility. It is intended to address the core features that bridge the indicated gaps while taking into account the time constraints of this research.

1.12.1 In-scope

The current research will include:

- **ECNN model development and training**: Designing and implementing the model based on equivariant convolutional neural networks for medical anomaly detection.
- **Use of MRI-based brain tumour datasets**: Employing publicly available datasets like BraTS for training and validation.
- **Evaluation against baseline models**: Comparing model performance with traditional CNN and related architectures to assess improvements in proposed model
- **Integration of explainability mechanisms**: Incorporating model interpretability tools and techniques to enhance clinical trust.

1.12.2 Out-scope

This research will have certain features excluded but added as future enhancements:

- **Clinical deployment or live patient testing**: The research does not extend to real-time clinical trials or integration within hospital systems.
- **Hardware optimisation or mobile adaptation**: Model compression or deployment on embedded systems is excluded.
- **Expansion to other organs or diseases**: While the model is designed to be generalizable, testing across organs other than the brain is outside the current scope.
- **Patient data acquisition**: Only publicly available datasets will be used, and no direct patient data collection will be performed.
- **Commercialization or product development**: This research focuses on research and prototype validation rather than market-ready solutions.

1.13 Hardware/software requirements

1.13.1 Hardware

Hardware requirement	Description
Primary device - ASUS Vivobook 15 X1504VA, 13th Gen i7 (Window OS)	Primary research environment for development and testing.
16 GB RAM	Ensures smooth model training and efficient multitasking.
512 GB SSD	Supports high-speed data access and storage for large MRI datasets.
GPU/ cloud	For high-performance computation during training or hyperparameter tuning.

Table 3. Hardware requirements

1.13.2 Software

Software requirement	Description
Operating system - Windows OS	The primary environment for development and testing, providing compatibility with required frameworks and hardware resources.
Python and relevant libraries	Core programming language used for implementing the model, data preprocessing, and analysis, with libraries such as PyTorch, TensorFlow, NumPy, and Pandas.
VS Code/ Google Collab	Integrated Development Environments (IDEs) used for efficient coding and debugging,
Git/Github	Version control tools used to manage code changes and maintain versions
Google Drive	For data backups and maintaining synchronized project files.
Google Docs, Slides	Documentation and presentation tools used for report writing, progress tracking, and preparing presentations.
Zotero	Reference management tool used to organize, cite, and manage research literature efficiently.

Table 4. Software requirements

1.14 Chapter summary

This chapter has presented a foundational understanding of the medical imaging domain where the problem exists, highlighting the significance and complexity of anomaly detection challenges. It clearly defined the research problem and motivation, establishing why innovative approaches are necessary. A thorough review of existing work emphasized current limitations and identified critical gaps that this research aims to address. The chapter outlined the specific contributions made to both the research and problem domains and discussed key challenges to be overcome. Finally, it posed focused research questions, stated the aim, and set clear objectives for the study. Together, these components provide an opening for the upcoming chapters.

CHAPTER 02: LITERATURE REVIEW

2.1 Chapter overview

This chapter provides a comprehensive review of the problem domain and existing literature related to deep anomaly detection in biomedical imaging, highlighting its methodological advancements, clinical importance, and challenges.It explores foundational and evolving deep learning architectures, including autoencoders, variational models, and hybrid CNN–Transformer networks, followed by an examination of transfer learning and geometry aware techniques and then analytically reviews on each stage in deep learning pipelines employed in previous studies. This chapter lays the conceptual and technological foundation required to inform and justify the proposed framework developed in later chapters by utilizing the findings of earlier studies and identifying persistent gaps.

2.2 Problem domain

2.2.1 Anomaly detection in the biomedical imaging domain

Anomaly detection in biomedical imaging is a fundamental task which attempts to detect deviations in the expected anatomical or physiological structures, which could indicate either the existence of disease or abnormality. Traditionally, this has been done using skilled visual examination by clinicians that could be time consuming and subject to inter-observer error. The recent growth of the volume of imaging data and the complexity of modalities has in turn fueled the creation of automated anomaly detection systems, especially ones that are based on deep learning (Fernando et al., 2020; Tschuchnig and Gadermayr, 2021).

DL models, such as autoencoders, generative adversarial networks (GANs), and transformer-based architectures, have become central to this field due to their ability to learn non-linear and hierarchical representations of healthy anatomical patterns, thus enabling the detection of subtle or heterogeneous abnormalities (Lu et al., 2024; Pinaya et al., 2022; Lei et al., 2024). They are normally trained on large collections of healthy images, and learn to reconstruct or model the manifold of normal anatomical structures. The deviations of such learned distribution, which are measured by reconstruction error, latent-space distance, or latent-space abnormality score, are used as interpretations of possible anomalies (Baur et al., 2020; Pinaya et al., 2022; Luo et al., 2023). It is especially useful in areas of clinical imaging

where abnormal data are limited and heterogeneous and are expensive to annotate, and supervised classification methods are not feasible (Nawaz et al., 2024).

Unsupervised and self-supervised anomaly detection models have been shown to identify tumours, lesions and degenerative tissue changes in brain MRI by identifying areas that do not match the reconstructed model of normal brain structure. In addition to neuroimaging, neuroimaging has shown that the same principle can be effectively applied to other modalities, including retinal Optical Coherence Tomography (OCT), chest X-rays and histopathology, thus, exhibiting the versatility and clinical potential of anomaly detection for its broad applicability in biomedical imaging (Lei et al., 2024; Lu et al., 2024).

2.2.2 Challenges and significance of anomaly detection in biomedical imaging

Despite its promise, anomaly detection in biomedical imaging faces several persistent challenges. One of the most significant is the scarcity of annotated abnormal data, as their acquisition and annotation are time-consuming, costly, and often not feasible in the case of infrequent diseases (Lu et al., 2024; Pinaya et al., 2022). This limitation is commonly recognised to be a major bottleneck to supervised learning methods where large, heterogeneous and well labelled datasets are required to achieve optimal performance. This has resulted in a shift toward unsupervised and weakly supervised learning approaches, based largely on normal images to train models. Even though such strategies minimize the use of labeled data, they introduce new challenges, in particular, the tendency to overfit, particularly when the diversity of normal data is insufficient or when models tend to learn trivial identity mappings rather than meaningful representations of normal anatomy(Lu et al., 2024; Baur et al., 2020; Lei et al., 2024).

To address the limited availability of annotated data, data-augmentation methods are systematically introduced to increase the amount and diversity of the data. However, these processes might not capture the complex heterogeneity of the real pathological manifestations, and in some cases, they can sometimes introduce artifacts that negatively impact model performance(Lu et al., 2024). The augmentation strategies, such as geometric transformations or synthetic lesion generation, do not typically correspond to the natural distribution of the pathological properties and trigger models that do not generalize to previously unobserved clinical cases (Ounasser et al., 2024). Also, the high levels of false-positives remain a relevant issue in clinical practice. Models are prone to false detection of benign anatomical deviation or imaging artefacts in the form of anomalies, thus resulting

in unnecessary follow-up research and burdening clinicians. This issue is particularly pronounced in unsupervised settings, where the lack of abnormal training examples can make it difficult for models to distinguish between clinically relevant anomalies and normal anatomical variability (Baur et al., 2020; Lei et al., 2024).

These challenges highlight the need for continued research into more robust, generalizable, and clinically applicable anomaly detection methods in biomedical imaging. Addressing the issues of data scarcity, overfitting, limited augmentation, and high false positive rates is essential for the development of reliable systems that can support clinicians in early and accurate disease detection.

The significance of advancing anomaly detection methods in biomedical imaging is substantial. Automated systems may also support clinical decision-making by providing objective, reproducible, and fast analyses, which are critical when dealing with high-stakes tasks such as the diagnosis of cerebral neoplasms (Luo et al., 2023). They have the potential to reduce the number of diagnostic errors, decrease the workload of medical professionals, and democratise access to expert-level diagnostics particularly under resource-limited settings (Pinaya et al., 2022). Therefore, it is of paramount importance that these gaps are addressed in order to come up with sufficiently strong, generalisable, and clinically applicable anomaly-detection systems that are able to improve patient outcomes and adapt to the evolving needs of the healthcare infrastructure. The proposed model is motivated precisely by these gaps by embedding spatial symmetry and transformation awareness directly into the model to enhance consistency, reduce augmentation dependency, and produce more trustworthy anomaly maps in brain MRI analysis.

2.3 Existing work

Deep learning has transformed anomaly detection in medical imaging and analysis, enabling automated identification of subtle and complex abnormalities across modalities. The subsequent sections review recent advances, highlighting the evolution of methods, key architectures, hybrid and transfer learning strategies, and the persistent gaps that motivate the present study. Given the vast and rapidly evolving nature of this domain spanning through diverse imaging modalities, architectures, and clinical applications, this review focuses on consolidating the most relevant developments that inform the research's aim of improving medical anomaly detection through advanced deep learning architectures for enhanced robustness and generalizability.

2.3.1 Core deep learning architectures

Classical deep learning paradigms, such as Autoencoders (AEs), Variational Autoencoders (VAEs) and , form the backbone for unsupervised anomaly detection in many medical imaging applications, by learning to reconstruct normal anatomy and flagging deviations as anomalies. Existing work include, Yun et al. (2023) where they demonstrated that the Sliced-Wasserstein Autoencoder outperformed traditional AE and VAE models in breast ultrasonography, addressing challenges related to image quality variability, operator dependency, and class imbalance. However, though it outperformed traditional AE/VAE models in breast ultrasonography, reducing these false positives remains a crucial challenge for subsequent studies aiming for clinical deployment.

Shvetsova et al. (2020) tackled the detection of subtle abnormalities in high-resolution chest X-rays and pathology slides by proposing a deep perceptual autoencoder with a redesigned training pipeline. This method introduced a robust abnormality scoring approach and relaxed the fully unsupervised assumption by using a minimal set of anomalies for hyperparameter initialization, establishing a new baseline for anomaly detection in complex medical imaging tasks. However, this approach still depends heavily on handcrafted hyperparameter tuning, limiting scalability across datasets. Moverover, Zhang, H et al. (2022) proposed an improved adversarial autoencoder incorporating a chain of convolutional blocks (CCB) to preserve global and local information while reducing semantic disparity between encoder and decoder features. This approach minimized reconstruction error in both image and latent vector spaces, achieving superior performance across multiple modalities including breast, retinal, lung, and skin imaging.

Moreover, multiple studies including a review by D. Sindhura et al.(2024), shows how Generative Adversarial Networks (GANs) have been applied for anomaly detection in a tabular format, offering detailed anomaly maps and improved performance in weakly supervised settings. However, the review highlights key limitations, including training instability, mode collapse, and high false-positive rates in complex or heterogeneous data. While GANs enhance visual localization and handle data scarcity, their lack of stability and interpretability limits consistent clinical deployment.

Comparative studies have further informed methodological development. Baur et al. (2020) conducted a systematic comparison of unsupervised autoencoder-based anomaly detection approaches for brain MRI, standardizing architecture, resolution, and datasets to improve

comparability and evaluate domain shift sensitivity. Despite their systematic design, Baur et al. identified critical limitations, including overfitting, limited generalizability across institutions and imaging protocols, and high false positive rates arising from the subtlety of anomalies and data imbalance inherent to medical datasets. These findings underscored that even carefully regularized models tend to memorize rather than generalize, challenging the reproducibility of anomaly detection systems across clinical environments.

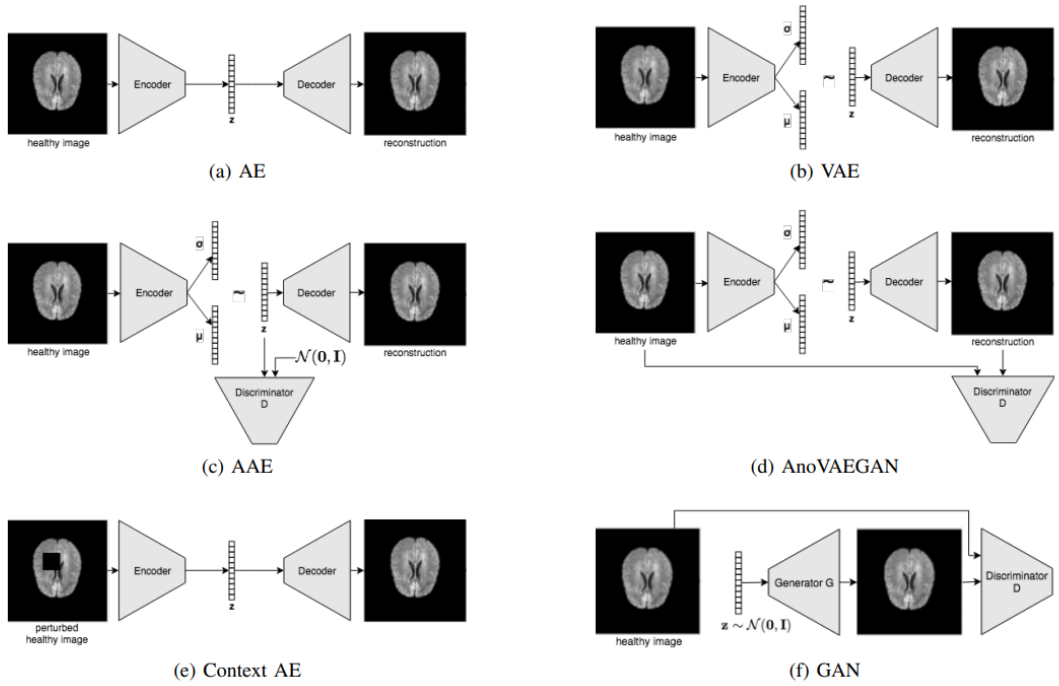


Figure 1. Autoencoder-based architectures for Unsupervised Anomaly Detection at a glance. Baur et al. (2020)

2.3.2 Existing hybrid and transfer learning approaches

Apart from the traditional and core techniques, recent advances increasingly rely on hybrid architectures and transfer learning to address limitations of pure CNNs or Transformers and to mitigate challenges posed by limited annotated datasets. A comprehensive survey on deep learning advancements in Anomaly Detection by Huang et al. (2025) discusses the increasing importance of hybrid approaches. In the context of AD, hybrid methods involve the integration of traditional and deep learning methods. This integration is advantageous because it combines the interpretability of traditional techniques with the flexibility of deep learning, ultimately working to enhance detection accuracy and model transparency in complex data distributions. This dual benefit addresses a critical challenge for DNNs in MIA, which often lack transparency in clinical workflows.

Hybrid autoencoder models have also emerged for unsupervised anomaly detection. Lu et al. (2024) proposed the Heterogeneous Autoencoder (Hetero-AE), combining a CNN encoder with a CNN-Transformer decoder, augmented with multi-scale sparse transformer blocks and multi-stage feature comparison. This architecture mitigates overfitting and improves anomaly detection across multiple modalities, including retinal OCT, chest X-ray, brain MRI, and COVID-19 imaging, while providing interpretable heatmaps. Sufian and Niu (2025) developed a hybrid AE-CNN-RNN framework for cardiac MRI segmentation, demonstrating improvements in structural analysis, early diagnosis accuracy, and patient outcomes.

Hybrid CNN-Transformer networks, for example, combine the local feature extraction capability of CNNs with the global context modeling of Transformers, achieving improved segmentation accuracy and generalizability. The PFormer network (Gao et al., 2025) contributes to the established trend of CNN-Transformer hybrid architectures by designing an efficient model specifically for 3D medical image segmentation. Its reliance on a specialized "content-driven P-attention" mechanism aims to efficiently integrate CNN's local feature extraction with the Transformer's long-range dependency modeling, ultimately achieving more precise segmentation while reducing computational burden, while a review on brain tumour detection with deep learning and hybrid approaches by C. Sarada et al. (2025) reviews recent hybrid architectures for brain tumour detection in MRI, particularly combinations like CNN-SVM and CNN-RNN models. These approaches merge CNN’s strong feature extraction with SVM’s precise classification or RNN’s temporal pattern analysis, achieving high accuracy (often > 98 %) in tumour segmentation and classification. However, the study notes key limitations such as overfitting, limited generalisability across datasets, and high computational complexity, highlighting the need for more robust and clinically adaptable models.

Liu et al. (2023) proposed STA-Former, a hybrid model incorporating a hierarchical encoder, a Shrinkage Triplet Attention module in the decoder, and multiscale attention-based feature fusion. This architecture effectively captures both short- and long-range dependencies and outperforms state-of-the-art methods across multiorgan CT, cardiac MRI, and breast ultrasound datasets. Li et al. (2025) introduced a Dual-Encoder Fusion Model that integrates CNNs for local context modeling with Transformers for long-range dependency capture, using a Correlation Fusion Module to optimize the integration. The model demonstrated notable improvements on the Synapse dataset and outperformed fine-tuned nnU-Net

baselines on tumour detection tasks in MSD datasets. Additionally, in the domain of 3D MRI, Anuradha et al. (2024) applied ConvLSTM integrated with U-Net architectures to achieve accurate anomaly detection with high sensitivity, demonstrating the potential of recurrent components in capturing temporal and spatial dependencies in volumetric data.

For 3D imaging, Labbihi et al. (2024) presented a hybrid 3D segmentation network combining CNNs with a frequency transformer branch. This approach leverages volumetric spatial features extracted by the CNN encoder and processed through a Fourier-transform-based transformer, achieving remarkable Dice scores for kidney, tumour, and brain datasets. Gao et al. (2025) introduced PFormer, an efficient CNN-Transformer hybrid using content-driven P-attention to improve segmentation precision while reducing computational cost. Similarly, HCTNet and AFC-Unet (He et al., 2023; Meng et al., 2025) demonstrate how attention mechanisms and multi-scale feature fusion in hybrid architectures can enhance anomaly localization and reduce semantic discrepancies, particularly in breast ultrasound imaging.

Transfer learning further enhances label efficiency and domain adaptability. Li et al. (2020) proposed a joint optimization approach for fine-tuning pre-trained models on small target datasets, yielding superior discriminative performance for challenging anomalies. Alloqmani et al. (2023) implemented a two-stage framework using MobileNetV2 for feature extraction followed by a perceptron classifier, achieving high accuracy for breast cancer anomaly detection while effectively handling data imbalance. However, while transfer learning provides strong performance gains and is essential for addressing issues like imbalanced data, it implicitly relies on the initial, often massive, pre-training dataset to establish feature knowledge.

2.3.3 Advances of deep learning techniques over the years

Deep learning has revolutionized anomaly detection (AD) in medical image analysis by automating the identification of subtle and complex pathologies across modalities such as MRI, CT, and X-ray. Conventional approaches relied on manual feature extraction, but the emergence of Convolutional Neural Networks (CNNs) enabled end-to-end learning for classification and segmentation tasks (Zhou et al., 2020; Sistaninejhad et al., 2023). Studies such as Kumar et al. (2025) reported diagnostic accuracies exceeding 95%, demonstrating the superiority of deep learning over traditional techniques. Sequential models, including RNNs

and LSTMs, further extended these capabilities to capture spatiotemporal dependencies, particularly in 3D and volumetric brain MRI anomaly detection (Ramtekkar et al., 2023).

Early in the past five years, research emphasized unsupervised and generative frameworks to detect rare or subtle abnormalities. Ruff et al. (2020) highlighted the transition toward reconstruction- and generation-based approaches, while Generative Adversarial Networks (GANs) emerged as key tools to address data imbalance and limited annotations through synthetic data generation (Sindhura et al., 2024). During this period, hybrid and multi-component 3D MRI models gained traction, improving volumetric anomaly localization (Bindu and Devi, 2024).

Subsequently, research focused on more advanced unsupervised 3D methods and hybrid architectures for improved generalization and interpretability. Pinaya et al. (2022) combined vector quantized variational autoencoders with autoregressive Transformers for volumetric brain anomaly detection, achieving high performance without complex post-processing. Similarly, Zhao et al. (2021) introduced the SALAD framework, employing self-supervised and translation-consistent features alongside structure similarity and center constraint losses, demonstrating effectiveness on OCT and chest X-ray datasets. A comprehensive survey by Huang et al. (2025) emphasized the importance of hybrid approaches, combining traditional and deep learning elements to enhance detection accuracy and model transparency. Concurrently, advanced architectures like Vision Transformers (ViT) and Ensemble Learning (EL) are increasingly explored for cancer diagnosis in medical images (Jiang et al., Jul 1, 2023).

Despite these advancements, ongoing challenges remain, including CNNs' difficulty capturing spatial relationships, the need for extensive data augmentation for 3D invariance, scarcity of labeled medical images, slow and costly imaging technologies, and persistent issues with generalization, false positives, and label efficiency. Collectively, these developments indicate a shift toward models that are mathematically robust, multimodally adaptable, clinically interpretable, and designed to assist rather than replace medical professionals.

A simplified timeline of key approaches, techniques, and developments in deep learning-based anomaly detection for medical imaging in a tabular format can be found in **Appendix A.1.**

2.4 Technological review

The subsequent sections present a critical analysis on the technical review on each stage of the engineering pipeline through various studies done by researchers.

2.4.1 Data Acquisition, pre-processing and feature extraction

2.4.1.1 Data sources and acquisition

Recent research in medical image anomaly detection spans a wide spectrum of imaging modalities and data sources prominently relying on publicly available brain MRI datasets such as BraTS, IXI, and Medical Segmentation Decathlon (MSD) for both supervised and unsupervised learning. The BraTS dataset provides high-quality, multi-institutional volumetric MR images with four modalities—T1, T1CE, T2, and FLAIR—annotated for glioma segmentation, making it an essential reference for evaluating anomaly detection and segmentation algorithms (Anuradha et al., 2024; Pinaya et al., 2022). The IXI dataset, in contrast, offers healthy brain scans and serves as a critical baseline for unsupervised anomaly detection studies, particularly when used alongside pathological datasets such as BraTS or MSSEG2015 (Luo et al., 2023). Complementary datasets like DMS, DGB, and MSLUB provide multiple sclerosis and glioblastoma lesions, enabling models such as autoencoders and equivariant networks to generalize across diverse pathologies (Baur et al., 2020).

Beyond Brain MRI, several other modalities have contributed significantly to the development of deep learning-based anomaly detection architectures. Retinal OCT datasets, including OCT2017 and ACRIMA, have facilitated unsupervised lesion detection and retinal layer segmentation using autoencoder and transformer-based methods (Lu et al., 2024). Similarly, BUSI and BUS datasets have supported segmentation research in breast ultrasonography (He et al., 2023; Yun et al., 2023), while Synapse and ACDC datasets have advanced hybrid CNN-transformer architectures for multi-organ and cardiac MRI segmentation (Liu et al., 2024; Gao et al., 2025).

2.4.1.2 Pre-processing and feature extraction techniques

Preprocessing acts as the foundation for ensuring data uniformity across subjects and modalities.To standardize input volumes, commonly used techniques include resampling, intensity normalization, bias field correction, and artifact removal. For example, Auradha et al. (2024) normalized MRI volumes to isotropic voxel resolutions for volumetric segmentation, whereas Yun et al. (2023) used Gaussian filtering, gamma correction, and

normalization in ultrasonic imaging. For CNN and Transformer architecture compatibility, images in 2D-based frameworks are usually downsized to 224×224 or 256×256 pixels (Meng et al., 2025; He et al., 2023). Patch-based resampling is commonly used for volumetric scans in order to overcome memory limitations in 3D CNNs (Labbihi et al., 2025).

Furthermore, under restricted data regimes, **data augmentation** is still a crucial method for improving generalization. Elastic deformation, rotation, flipping, and intensity perturbation are examples of common transformations (Meng et al., 2025; Gao et al., 2025). However, excessive augmentation can compromise segmentation fidelity by distorting biological features, particularly in vital areas like the brain. In the meantime, deep learned embeddings that capture intricate spatial and contextual dependencies have replaced traditional handcrafted methods for feature extraction and representation learning, such as texture, shape, and intensity descriptors. While more recent designs use attention mechanisms for feature refinement, reconstruction-based models (e.g., Baur et al., 2021; Yun et al., 2023) use residual errors between input and reconstructed images to identify abnormalities. For instance, Meng et al. (2025) introduced a Multi-Scale Fusion Attention Gate (MFAG) for enhanced boundary localization, and Liu et al. (2024) applied Shrinkage Triplet Attention (STA) to integrate spatial and channel-wise information efficiently.

Despite advances in preprocessing and feature extraction for medical image anomaly detection, several limitations persist. Model generalization may be impacted by preprocessing pipelines' discrepancies in intensity normalization, resizing, and artifact removal due to their frequent lack of consistency across datasets and modalities (Yun et al., 2023; Anuradha et al., 2024). Although it increases robustness, extensive data augmentation can compromise precision in minor or mild lesions and distort anatomical structures, also used in standard CNNs and hybrid CNN–Transformer models which increases computational cost and may introduce bias. Clinical interpretation of feature extraction techniques, especially deep-learned embeddings, is often challenging due to their opaque representations (Baur et al., 2021; Pinaya et al., 2022). Moreover, reconstruction-based anomaly detection approaches are limited by their dependence on accurately modeling healthy anatomy, making them less effective when anatomical variability is high or datasets are small. These challenges highlight the need for preprocessing and feature extraction methods that are standardized, interpretable, and robust to geometric transformations, motivating the development of the proposed ECNN framework.

2.4.2 Deep Learning Architectures

2.4.2.1 Autoencoder based architectures

Autoencoders (AEs) remain a core architecture for unsupervised anomaly detection in medical imaging, primarily because of their capacity to learn compact latent representations of healthy anatomy. This framework interprets differences between input and reconstructed outputs as anomalies. The network reconstructs the input picture while minimizing a reconstruction loss function $L=||x-x^||2$. For brain MRI anomaly segmentation, foundational research by Baur et al. (2020) and Luo et al. (2023) thoroughly assessed classical and variational autoencoders. They found that although reconstruction-based techniques are effective in highlighting abnormal regions, they struggle to capture global anatomical consistency.

A number of extensions have tried to improve these restrictions. To increase robustness and visual accuracy, models like f-AnoGAN and AnoVAEGAN, as well as variational autoencoders (VAEs) and adversarial AEs (AAEs), add probabilistic priors or adversarial loss terms. While Lu et al. (2023) presented the Dual-Encoder Autoencoder (DE-AE), which incorporates auxiliary encoders and transformer layers for multi-scale feature enrichment, Javed et al. (2024) proposed an AECNN (Auto-Encoder CNN) using a symmetric encoder–decoder with convolutional and transpose-convolutional layers. Similarly, C-SAN, a convolutional stacked autoencoder that integrates deep feature hierarchies for improved localization of brain tumours, was proposed by Gayathiri & Santhanam (2025).

Despite these improvements, conventional autoencoders remain prone to overfitting on small datasets and are geometrically naive, meaning they cannot inherently handle variations in orientation or scale, which is a significant issue in 3D MRI scans with arbitrary patient positioning.

2.4.2.2 Hybrid CNN–transformer and attention models

As transformer designs have become more popular, a number of studies have explored hybrid CNN–Transformer networks, which combine the global context modeling of self-attention processes with the spatial locality of convolutions. AFC-Unet is a full-scale attention-fusion model created by Meng et al. (2025) that combines hierarchical feature maps across scales for accurate segmentation. Similarly, He et al. (2023) proposed HCTNet, which successfully

maintains a balance between contextual reasoning and local feature sensitivity by utilizing Transformer Encoder Blocks (TEBs) alongside Spatial–Channel Attention (SCA) modules.

While Liu et al. (2024)'s STA-Former introduces Shrinkage Triplet Attention (STA) for stable convergence and improved focus on pertinent structures, the SALAD framework (Self-supervised and Translation-consistent Anomaly Detection) improves anomaly localization without labels by enforcing translation consistency between input and reconstructed feature maps.

This integration is extended to volumetric data by other hybrid models as Labbihi et al. (2025)'s 3D Frequency Transformer Fusion architecture and PFormer (Gao et al., 2025), which combine CNN feature extractors with frequency-domain transformer blocks for better 3D segmentation. Despite their great accuracy, these models have substantial training costs and memory footprints. To prevent overfitting, they frequently require massive data augmentation and pre-trained backbones (such as ResNet50 or Swin Transformer). Thus, transformer-based techniques enhance representation richness, but for small, domain-specific datasets, they are data-inefficient and computationally prohibitive.

2.4.2.3 Geometry-Aware and Equivariant Architectures

A more recent, mathematically based approach to medical imaging is represented by geometry-aware architectures, which are made to specifically maintain rotational and reflectional symmetry in neural representations. These models use group-equivariant convolutions, in which filters are required to change predictably under a specific set of transformations (such as flips and rotations).

Group Equivariant CNNs (G-CNNs) were pioneered in lung CT and histopathology by Winkels & Cohen (2019) and Lafarge et al. (2020), respectively. They used rotation groups such D4, D4h, and SE(2,N) to preserve consistent feature maps across orientations. Pang et al. (2022) created Group Equivariant Res-UNet (GER-UNet) to incorporate equivariance into U-Net backbones, enhancing rotational resilience in segmentation tasks, while Yan Li et al. (2020) extended this idea to 3D data through Dynamic Group Equivariant DenseNets.
Even while these models greatly increase rotational invariance and decrease overfitting, they are still computationally costly because each convolutional operation needs to compute several modified versions of a filter. Furthermore, parameterization and group selection are still challenging, limiting widespread adoption.

By combining autoencoder simplicity and geometric consistency, the proposed ECNN based model aims to bridge the gap. The model may develop symmetry-preserving latent representations that remain stable under transformations by embedding group-equivariant convolutions within an unsupervised AE framework—without the need for transformer pretraining or extensive data augmentation. This method offers a lightweight, geometry-aware substitute, directly addressing the drawbacks of hybrid CNN–Transformers (data inefficiency) and traditional AEs (geometric blindness).

A table summarizing the comparison of major architectures for medical image anomaly detection, can be found in **Appendix A.2.**

2.5 Evaluation and Benchmarking

2.5.1 Evaluation Strategies and Metrics

Evaluation strategies that have evolved in medical image anomaly detection and segmentation include both region-wise lesion detection accuracy and pixel-level reconstruction quality. Reconstruction based measures, particularly Root Mean Squared Error (RMSE), Mean Absolute Error (ℓ_1), and Dice Similarity Coefficient (DSC), are emphasized in studies like Yun et al. (2023) and Baur et al. (2020) to evaluate how closely the rebuilt image resembles the healthy anatomy. These measure how the input image x and its reconstruction x differ at the pixel level:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(x_i - \widehat{x_i})^2 , \quad RMSE = \sqrt{MSE}$$

For segmentation tasks, the Dice coefficient remains the dominant metric, providing region-level overlap accuracy. Sensitivity (SEN), Specificity (SPE), Accuracy (ACC), and F1-score are widely used across Meng et al. (2025), He et al. (2023), and Gao et al. (2025) to quantify the precision–recall balance and clinical reliability of predictions. Given below are the mathematical formulas to calculate the specific metric, where TP is true positives, FP is false positives, TN is true negatives and FN is false negatives;

$$Dice\ coefficient: \frac{2 * TP}{(TP+FP) + (TP+FN)} , \quad Accuracy: \frac{TP + TN}{TP + TN + FP + FN} ,$$

$$Sensitivity: \frac{TP}{TP+FN} , \quad Specificity: \frac{TN}{TN+FP}$$

Recent studies, like those by Pinaya et al. (2022), Lu et al. (2024), and Tang et al. (2025), have focused on threshold-independent metrics like Area Under the Receiver Operating Characteristic Curve (AUROC) and Area Under the Precision-Recall Curve (AUPRC). AUPRC concentrates on performance under class imbalance (e.g., small lesion areas), whereas AUROC assesses the trade-off between true positive and false positive rates. These are essential, since there are few ground truth lesion boundaries for unsupervised anomaly detection. Furthermore, segmentation studies are increasingly using Hausdorff Distance (HD) and its variations (HD95, ASSD) to evaluate boundary accuracy (Liu et al., 2024; Labbihi et al., 2025).

2.5.2 Benchmarking strategies

Benchmarking in medical image analysis typically involves utilizing consistent training and data sets to compare new models with defined baselines. Classical baselines include autoencoders (AE) and variational autoencoders (VAE) (Yun et al., 2023; Baur et al., 2020). In order to assure fair comparison, it usually involves evaluating model performance against well-known public datasets using specified measures. In order to evaluate generalization, researchers frequently employ a fixed train and test split, in which models are trained only on healthy samples and evaluated on both normal and unhealthy instances.

Generative variations such as f-AnoGAN, GANomaly, and SALAD expand these baselines by enhancing anomaly localization through self-supervised and adversarial methods (Lu et al., 2024; Tang et al., 2025). Examples of hybrid CNN–Transformer models that benchmark performance across a wide range of segmentation frameworks are such as U-Net, TransUNet, SwinUNet, and nnU-Net include AFC-UNet (Meng et al., 2025), HCTNet (He et al., 2023), and STA-Former (Liu et al., 2024). These models show consistent improvements in DSC and F1 while improving attention to fine structures.

In order to verify robustness and generalizability, benchmarking approaches have evolved by integrating multi-metric evaluation and cross-dataset validation. For instance, to quantify boundary precision and anomaly discriminating ability, hybrid CNN–Transformer models provide Hausdorff Distance (HD) and Area Under the ROC Curve (AUC) in addition to DSC and SSIM. To verify geometric consistency, equivariant architectures such as G-CNNs (Winkels & Cohen, 2019) and Dy-GCNNs (Li et al., 2020) benchmark their models on rotation-augmented test sets. This approach is in line with clinical requirements where brain

MRIs are recorded in a variety of orientations. However, because there are no standardized protocols or common leaderboards, these comparisons are frequently fragmented among studies. The proposed ECNN based framework intends to adopt a multi-metric, cross-validation benchmarking strategy to quantify both reconstruction accuracy and geometric robustness, ensuring direct comparability with current SOTAs while highlighting its computational efficiency and symmetry-preserving advantages.

2.6 Concept map

A concept map visualizing the key concepts and methods within the domain of deep anomaly detection in biomedical imaging can be found in **Appendix A.3.**

2.7 Conclusion

In conclusion, the problem domain, the evolution of anomaly detection methodologies, tracing progress from traditional autoencoder-based reconstruction models to advanced hybrid and geometry-aware architectures and their implementation in biomedical imaging, specifically in brain MRI analysis, have all been thoroughly examined in this literature review. It reviewed how deep learning approaches have greatly improved the ability to identify subtle and complicated anomalies, but it also outlined ongoing challenges like model generalization, overfitting, reliance on large datasets and data augmentation, and robustness across various studies. These findings serve as the basis for the proposed deep anomaly detection framework covered in the upcoming chapters and directly motivate the research directions taken in this study, which aims to unify geometric consistency, lightweight design, and improved anomaly localization for more accurate and generalizable medical image analysis.

CHAPTER 03: METHODOLOGY

3.1 Chapter overview

The following chapter outlines how the research will be executed. It is structured into three main sections, including Research Methodology, Development Methodology, and Project Management Methodology. Each of these sections outlines the systematic approach along with justifications. The scope of the research is clearly stated as well. Further through the chapter, the grant chart, the deliverables, the system requirements and a detailed risk assessment and mitigation plan can be found.

3.2 Research methodology

Shown below is the chosen research methodology, structured following the Research Onion model proposed by Saunders et al. (2019).

Layer	Choice	Justification
Philosophy	Pragmatism	A pragmatic approach is appropriate because it ensures that the research findings meet real-world healthcare concerns by combining qualitative considerations of interpretability and clinical applicability with quantitative evaluation of model performance.
Approach	Hybrid method	Starting with deductive reasoning based on well-established theories in deep learning, anomaly detection, and equivariance, a mixed-method approach also incorporates inductive reasoning to extract new insights from executions on medical imaging datasets. Through this two-step procedure, the author can refine novel contributions to the research domain while testing theoretical hypotheses.
Methodological Choice	Mixed method	A mixed-method underpins the methodology, with quantitative analysis forming the core through experimental evaluation of anomaly detection models, complemented by qualitative assessments

		of clinical relevance through interviews and feedback, strengthening applicability.
Strategy	Experimentation Interviews	The research utilises multiple strategies like experiments that will be used to design, train and benchmark the proposed framework , alongside interviews with experts to gather insights and feedback, validate and assess the model outputs.
Time Horizon	Cross sectional	A cross sectional time horizon is employed with the use of publicly available datasets, so that the models may be developed, trained and validated without the need for lengthy clinical trials.
Data Collection	Secondary data	Data is sourced from publicly available, annotated medical imaging datasets such as BraTS, with potential integration of supplementary metadata where feasible.

Table 5. Research methodology

3.3 Development methodology

3.3.1 Requirement elicitation methodology

Requirement elicitation methodology is the process of gathering, obtaining and outlining the requirements and specifications of the research. In order to elicit requirements, various techniques are used. Some of the best-known approaches that have been selected to be used in moving forward with the research are as follows:

Literature review: A comprehensive review of the problem domain and existing work has been conducted to identify the current limitations and trends in biomedical anomaly detection, particularly in brain MRI analysis. These insights help to find gaps and validate comparisons between the current solutions and the proposed model, and guide the identification of core technical requirements for research and development.

Interviews: To gather qualitative information about clinical and technical expectations, semi-structured interviews will be held with experts in the biomedical and technical fields. While ensuring compliance with ethical and clinical standards, these discussions will help improve the model's design requirements, particularly its accuracy, interpretability, and feasibility in actual diagnostic settings.

Prototyping: An iterative prototyping approach will be employed to transform theoretical concepts into experimental models. Early-stage prototypes of the proposed architecture will be developed and tested to evaluate key performance indicators such as anomaly detection accuracy, false positive rate, and interpretability. Feedback from initial evaluations will inform subsequent refinements in architecture and data integration.

3.3.2 Design methodology

SSADM follows a linear, process-driven approach that emphasizes data flow and functional decomposition, making it suitable for systems with clearly defined requirements while, OOADM adopts a modular and iterative framework that models a system as a collection of interacting objects, each encapsulating data and behavior, making it particularly effective for complex, adaptive, and research-based software systems such as those involving artificial intelligence and deep learning.

Given the iterative and modular nature of the proposed framework, which integrates components such as data preprocessing, equivariant graph construction, model training, and performance evaluation, the OOADM approach is adopted for this research. OOADM enables the design to be structured around key system objects, ensuring flexibility, scalability, and clear abstraction between components. Moreover, its emphasis on reusability and modularity aligns well with the need for experimental refinement in deep learning workflows.

3.3.3 Programming paradigm

The proposed research adopts the Object-Oriented Programming (OOP) paradigm to support a modular, scalable, and maintainable system design. OOP enables the encapsulation of data and functionality within objects, promoting reusability and abstraction across different components of the framework. In the context of the proposed model, this paradigm allows the seamless integration of modules for data preprocessing, graph construction, model training, and evaluation as separate but compatible entities. The implementation will primarily utilise Python, including the deep learning libraries such as PyTorch and TensorFlow, and other libraries for geometric deep learning and medical image analysis. This approach ensures flexibility for iterative experimentation and clear separation of concerns, enhancing both code readability and extensibility throughout the research.

3.3.4 Testing methodology

Model testing involves rigorous evaluation of each developed deep learning model to quantify its effectiveness in detecting anomalies in medical images. Some of the metrics and methodologies that will be utilised include:

- Accuracy, Precision, Recall, F1-Score
- Area Under Curve (AUC) - ROC and PR Curves
- Explainability Metrics: SHAP values or grad-CAM outputs.
- Benchmarking: Performance will be compared against state-of-the-art models, including CNN baselines and existing equivariant or multimodal fusion architectures to establish relative improvements.

In addition, feedback from experts will help qualitatively evaluate the proposed model in the testing phase.

3.3.5 Solution methodology

The implementation pipeline for the proposed model consists of several critical steps, each designed to address key challenges such as limited data, spatial variability, and multimodal integration.

Dataset collection: The initial phase involves the collection of diverse brain MRI datasets paired with relevant clinical metadata. Publicly available datasets such as BraTS and IXI will be primarily utilised due to their wide acceptance and extensive annotations, enabling robust benchmarking against prior work.

Data pre-processing: Preprocessing will include standardized techniques ensuring data consistency, normalization, and structural readiness for geometric deep learning. Due to the key role of spatial equivariance in the proposed approach, preprocessing will specifically preserve anatomical geometries critical for symmetry-aware learning instead of relying on extensive data augmentation. However, minimal augmentation may be applied solely for baseline comparison, ensuring fairness when evaluating traditional models if necessary.

Model selection: The research will focus on the Equivariant Convolutional Neural Network (ECNN) architecture, enhanced with symmetry-aware learning mechanisms that capture rotational and translational invariance inherent in biomedical structures. The selection of ECNN is motivated by its ability to model geometric relationships directly in the feature space, reducing dependence on large labelled datasets and improving robustness to spatial transformations. Multimodal fusion layers capable of dynamically integrating image features with metadata will be incorporated to enhance diagnostic accuracy as another module. Explainability modules like SHAP or Grad-CAM will be integrated for the interpretability module. In addition, baseline models will include standard CNNs to benchmark improvements.

Model training: The model will be trained using a supervised approach on the availability of labelled data. Training will involve optimising loss functions that balance reconstruction accuracy and anomaly localisation performance, along with regularisation techniques and adaptive optimisers to prevent overfitting.

Testing: Testing will include evaluation metrics such as accuracy, precision, recall, F1-score, and AUC for ROC/PR curves. Interpretability evaluation will assess the clinical relevance of generated explanations through expert review. Model generalisation will be validated on external datasets from different institutions or scanners. Moreover, comparative analysis with baseline models will determine the improvement achieved through the incorporation of symmetry-aware mechanisms and grid-based learning.

Feedback loop: A critical feedback loop will iteratively refine the model and system based on testing outcomes. Loopholes identified during testing will guide targeted improvements in architecture, training strategies, or data handling. Continuous collaboration with experts will ensure the solution remains aligned with diagnostic relevance.

3.4 Project management methodology

This research will be adopted on the *PRINCE2 Agile* methodology, which combines the flexibility and responsiveness of Agile approaches along with the structured process of PRINCE2. This hybrid approach makes it ideal for the experimental nature of the project, with constant improvements after evaluations

3.4.1 Schedule

3.4.1.1 Gantt chart

An updated Gantt chart with key milestones to date is included in **Appendix B.1.**

3.4.1.2 Deliverables

The following figure provides an overview of the deliverables along with their deadlines.

Delivarebles	Dates
Project Proposal	
Literature Review	
Project Proposal	10th Oct 2025
Project Proposal and Requirement Specification – final draft	
Project Proposal and Requirement Specification – Final	06th Nov 2025
Design Document	
Prototype	2nd Feb 2026
Interim Project Demo	2nd Feb 2026
Implementation	2nd Feb 2026
Testing	
Evaluation	
Thesis Submission	30th Mar 2026
Prototype	30th Mar 2026
Dataset and Source Code	30th Mar 2026

Figure 2. Project deliverables

3.4.2 Resource requirements

3.4.2.1 Data

Data requirement	Description
MRI dataset	This research will mainly rely on: 1. '**IXI**' dataset, which is one of the most widely used open source datasets in brain tumour imaging research. It contains 600 MR images from normal, healthy subjects (Anon, n.d.). 2. '**BraTS**' dataset, which is yet another very commonly used dataset. The dataset contains over 2000 manually labelled mpMRI brain scans from adult glioma patients and is specifically designed as a benchmark for tumour segmentation and classification using AI. (The, 2024) To enrich and enhance diagnostic accuracy by including diverse clinical perspectives, additional publicly available datasets will be acquired as necessary.

Table 6. Data requirements

3.4.3.2 Skills

Skill	Description
Programming - Python	Used for model implementation, preprocessing, and data handling.
ML and DL	Understanding algorithms, architectures, and evaluation metrics for model development and validation.
Research and documentation	Conducting literature reviews, managing references (Zotero), and writing reports.
Analytical and critical thinking	To interpret results, identify errors and optimize the model.

Table 7. Skills requirements

3.4.4 Risk and mitigation

Risk	Severity	Frequency	Mitigation Strategy
Dynamic Nature of Project Requirements.	4	4	Maintain proper change control documentation; use Agile principles to adapt iterations while preserving scope clarity.
System Failures or Poor Model Performance.	5	3	Use stable hardware, cloud backups, and efficient debugging tools; implement monitoring and performance benchmarking.
Lack of Domain Knowledge (Medical/Technical).	4	3	Conduct consultations with medical and AI experts; perform an extensive literature review and validation of results.

Unforeseen Issues (e.g., dataset bugs, environment errors).	4	3	Maintain flexible timelines; perform frequent saves, testing, and backups.

Table 8. Risk and mitigation

3.5 Chapter summary

Presenting an effective framework for carrying out the research, this chapter described the research, development, and project management methodologies. It covered the design and development processes, gathering and analysing information techniques, and management strategies that guarantee a project's timely and successful completion. The research is made both structured and robust by including specific requirements, explanations for methodological decisions, and risks that have been identified, along with appropriate mitigation techniques.

CHAPTER 04: SOFTWARE REQUIREMENT SPECIFICATION (SRS)

4.1 Chapter Overview

The subsequent chapter discusses the system's specifications in detail, including how they were elicited, the stakeholders involved, their viewpoints, and the functional and non-functional requirements of the system. The Rich Picture Diagram, Stakeholder Onion Model, Context Diagram, and Use Case Diagram are among the various illustrations that are included to help visualize how the system interacts. Later on, the MoSCoW approach is used to prioritize requirements, ensuring that the emphasis is on what is critical to the system's success.

4.2 Rich picture diagram

A visual representation of the system's high-level interactions with its users, stakeholders, and processes is given in the following diagram.

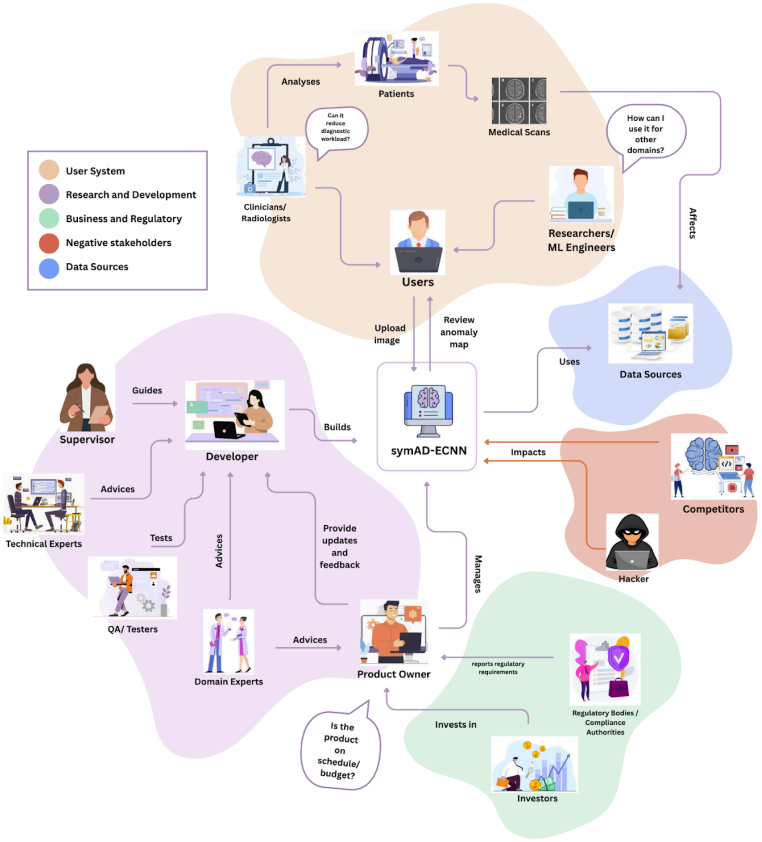


Figure.3. Rich picture diagram (self-composed)

4.3 Stakeholder analysis

The following section highlights the key stakeholders involved in the proposed system, highlighting their roles and interactions with the project. A stakeholder onion model illustrates the stakeholders' varying levels of engagement with the system. A detailed breakdown of each stakeholder, along with their contributions and impact on the development and functionality of the proposed system in research, can also be found in a tabular format below.

4.3.1 Stakeholder onion model

Figure 4. Stakeholder Onion Model (self-composed)

4.3.2 Stakeholder viewpoints

Stakeholder	Role	Description
Clinician / Radiologist	Normal user / Functional beneficiary	Uses the system/ model for better and easy medical image analysis.
Researcher /		Uses the system for further enhancements and

ML engineers		as baseline models for research.
Product owner	Operational beneficiary	Overlooks and ensures the system aligns with research goals.
Patients	Functional beneficiary	Receive validated insights from the system to assist diagnosis.
Developer	Service Provider	Develops and maintains the system/model.
Supervisor	Advisor	Guides and gives feedback on research progress.
Technical experts	Advisor	Guides and give insights on the technical aspects to make sure a smooth development process.
Domain experts	Advisor	Provides guidance and expertise on the domain related aspects.
Testers	Quality assurance	Tests and assures quality maintenance of the system/model.
Investors	Financial beneficiary	Invests in the project expecting financial gains
Regulatory/ compliance authorities	Regulator	Ensure the system adheres to regulatory compliance.
Competitors	Negative stakeholder	Develop alternate similar systems.
Hackers		Tries to get control of the system illegally.

Table 9. Stakeholder viewpoints

4.4 Selection of requirement elicitation methodologies

The selection of requirement elicitation methodologies is defined by the multidisciplinary nature of this study that incorporates deep learning tools with biomedical imaging modalities. A combination of techniques were chosen due to the technical complexity of the problem

domain. Such approaches ensure that the requirements derived are not just theoretically sound, but they are also practically proven.

Literature Review (LR)
The literature review approach provides a solid theoretical foundation and allows an in-depth exploration of literature to identify state-of-the-art methodologies, existing research gaps and limitations, and opportunities for innovation on which the suggested framework of deep biomedical anomaly detection is based. This ensures that the research objectives align with both academic and technological relevance.
Interviews
The interviews are used to complement the findings of the literature review by providing qualitative information of domain and technical researchers such as deep-learning and biomedical researchers and other relevant parties. These discussions on prevailing issues, model applicability, interpretability and feasibility thus supplement the exploration of the current study, and offer viable advice to facilitate a smooth project delivery.
Prototyping (Iterative Development)
Prototyping with the help of the iterative development process can allow testing of the hypotheses of the design systematically and testing of the suggested architecture. It enables early detection of design flaws, continuous refinement based on feedback, and performance benchmarking against key indicators like reconstruction error, anomaly localization accuracy, and computational efficiency.This approach bridges theory and practice by testing the proposed model on actual datasets.

Table 10. Requirements elicitation

4.5 Discussion of findings

4.5.1 Literature Review Findings

The key findings from an extensive literature review is presented along with the citations in the following table.

Key Findings	Citations
Anomaly detection in biomedical imaging identifies deviations in anatomical structures indicating disease. Manual inspection is time-consuming and prone to inter-observer variability. Deep-learning models such as AEs, GANs, and Transformers optimize representations across MRI, OCT, and X-ray and other modalities. However, limited annotated data restricts supervised learning and complicates distinguishing anomalies from normal variations.	Fernando et al., 2020; Lu et al., 2024; Baur et al., 2020; Lei et al., 2024
Autoencoders (AEs) and Variational Autoencoders (VAEs) trained on healthy data reconstruct normal anatomy to highlight deviations. Variants such as Sliced-Wasserstein AE, Deep Perceptual AE, and Adversarial AE show promise but overfit easily on small or homogeneous datasets, learning identity mappings instead of meaningful features.	Baur et al., 2020; Yun et al., 2023; Shvetsova et al., 2020; Lu et al., 2024
Hybrid architectures combining CNN and Transformer elements (HCTNet, STA-Former, AFC-Unet, PFormer) achieve high accuracy in tumor detection but suffer from low generalizability, large computational costs, and dependence on pre-trained backbones and extensive data augmentation.	Lu et al., 2024; Liu et al., 2023; Gao et al., 2025; Sarada et al., 2025; Huang et al., 2025
Data augmentation methods (rotation, flipping, elastic deformation, synthetic lesion generation) partially address small dataset issues but introduce artifacts and fail to capture real pathological diversity, reducing clinical reliability.	Lu et al., 2024; Ounasser et al., 2024; Zhou et al., 2020
Group-equivariant CNNs (G-CNNs, GER-Unets) leverage rotational and reflectional symmetries to improve robustness.	Winkels & Cohen, 2019; Pang et al.,

However, their computational cost and complex parameterization limit scalability. Conventional AEs remain inadequate for handling 3D MRI orientation and scale variations.	2022; Li et al., 2020; Baur et al., 2020
CNN-based methods achieve high accuracy and have evolved toward Vision Transformers and ensemble learning, yet still struggle with spatial reasoning, 3D invariance, and generalization, leading to false positives.	Zhou et al., 2020; Kumar et al., 2025; Pinaya et al., 2022; Huang et al., 2025
GANs generate synthetic data and detailed anomaly maps but are unstable during training (mode collapse), have high false-positive rates, and lack interpretability, limiting clinical integration.	Sindhura et al., 2024; Lu et al., 2024
Automated DL systems offer reproducible, objective diagnostic support and can reduce human diagnostic errors. However, black-box architectures, deep embeddings, and computational complexity hinder clinical acceptance.	Luo et al., 2023; Pinaya et al., 2022; Baur et al., 2020
Deep learning models tend to memorize rather than generalize, affecting reproducibility in clinical settings. High false-positive rates lead to unnecessary interventions and difficulty distinguishing benign anatomical variations from actual anomalies.	Baur et al., 2020; Lei et al., 2024; Lu et al., 2024

Table 11. LR findings

4.5.2 Interviews Findings

Initial unstructured interviews were conducted with the professionals who had significant experience in deep learning for medical imaging, both technical and domain-oriented, aiming to outline concrete issues, dataset considerations, and iterative model development cycles that inform the methodology of the study. Interviewees were selected through their technical expertise on deep learning and their domain knowledge in medical imaging, since this study primarily focuses on the technical challenges, such as data efficiency, geometrical understanding, and explainability rather than direct clinical application. Further interviews will be conducted throughout the study to continuously refine the research direction. A thematic analysis from the interviews are presented in the table below. Details of interviewees can be found in **Appendix C.1.**

Code	Theme	Conclusion / Key Insights
Domain Relevance	Biomedical Imaging Context	Interviewees emphasized that medical imaging anomaly detection, particularly in MRI-based brain tumor and lesion analysis, presents distinct challenges including those found in literature as their first hand experiences. Discussions revealed that MRI remains the most information-rich modality for structural anomalies, making it ideal for initial feasibility studies.
Data Imbalance and Generalization Limitations	Gaps in Current Research	Many professionals reported that small, imbalanced medical datasets cause overfitting and hinder generalization. Although augmentation and normalization are commonly suggested, these remain insufficiently optimized or standardized across anomaly detection studies. Hyperparameter tuning and robust validation strategies are often underexplored, creating inconsistencies in reported model performance.
Research Validation and Novelty	Addressing Identified Gaps	The interviews with professionals validated that the identified research gaps were not only theoretical but practically experienced in prior studies. Many reported instability in model generalization, limited access to large annotated MRI datasets, and challenges with black-box models. These recurrent difficulties confirmed that the gaps addressed by this research are real, experienced, and empirically grounded, not just literature-derived.
Dataset Feasibility and Preprocessing	Data Acquisition and Preparation	Experts discussed suitability of multiple imaging modalities (MRI, CT, Ultrasound) and highlighted the need for both healthy and abnormal samples for robust anomaly detection. They also emphasized converting

		3D MRI volumes into 2D slices to reduce computational complexity, and using batch processing pipelines for efficiency. This supports the project's decision to focus on brain MRI (BraTS, IXI) datasets with systematic preprocessing and slicing for manageable training.
Baseline and Iterative Development	Model Evolution Strategy	Interviewees described a structured workflow by beginning with simple CNN or AE baselines, followed by successive enhancements ultimately to equivariant convolutions. This iterative design cycle was considered essential for balancing interpretability, efficiency, and precision. The current study follows this philosophy by evolving from CNN-AE prototypes to an ECNN-AE architecture, demonstrating improvement.
Segmentation, Benchmarking, and Architectural Diversity	Model Design and Evaluation Framework	Professionals emphasized the importance of evaluating across multiple segmentation and reconstruction paradigms such as U-Net, VAE, and GAN-based architectures. Comparative benchmarking against traditional CNN and AE models was viewed as critical to ensure measurable improvement. Some also noted recent trends like diffusion models and attention mechanisms for finer segmentation and interpretability. These insights validate the research approach of systematic benchmarking to quantify the model's advantages in localization accuracy and generalization.
Explainability and Interpretability	XAI and Clinical Trust	Participants stressed that explainability is critical for medical imaging models. Integration of XAI tools (e.g., Grad-CAM, LIME, SHAP) and visualization libraries (e.g., Matplotlib, Plotly) were recommended for generating intuitive anomaly heatmaps. This

		justifies the research focus on explainable outputs to support clinical interpretability and build trust.
Libraries and Frameworks	Implementation Tools	Experts favored Python-based libraries particularly TensorFlow, PyTorch, and visualization libraries for XAI and analysis. Community-supported frameworks were preferred for flexibility and debugging ease. This supports adopting PyTorch for model development and XAI integration.
Hardware and Time Constraints	Practical Implementation Barriers	Researchers highlighted limited GPU resources and training time as key constraints. Cloud-based solutions such as Google Colab and Kaggle GPUs were commonly used to offset hardware limitations. This insight justifies the project's light prototyping focus on academic validation rather than full deployment.
Geometry aware and Advanced Techniques	Advanced Model Innovation	Interviewees with experience in geometry-based and equivariant models noted a steep learning curve but promising results in rotationally invariant medical data. This aligns with the project's innovation focus, contributing novel experimentation in equivariant feature learning.

Table 12. Interviews findings

4.6 Summary of findings

A summary of the overall findings from the requirements elicitation phase is given below.

ID	Finding	Literature Review	Interviews	Prototyping
1	Validate the research problem and identify existing gaps in biomedical anomaly detection using MRI.	X	X	
2	Evaluate and validate the novel approach.	X	X	

3	Identify key datasets (e.g., BraTS, IXI) suitable for MRI-based anomaly detection.	X	X	
4	Recognize challenges in dataset imbalance and the need for effective preprocessing and augmentation techniques.	X	X	X
5	Evaluate baseline architectures (CNN-AE, VAE, GAN-based) for performance comparison.	X	X	X
6	Highlight computational constraints and the need to optimize training efficiency.	X	X	X
7	Define the role of explainable AI (XAI) in improving explainability and clinical trust.	X	X	
8	Explore suitable XAI techniques (e.g., Grad-CAM, saliency maps) compatible with the ECNN-AE framework.	X	X	X
9	Emphasize benchmarking strategies and performance metrics for objective evaluation.	X	X	X
10	Identify iterative development as the most practical approach for model refinement and validation.		X	X
11	Determine feasibility of building a light prototype for demonstrating anomaly detection results.		X	X
12	Recognize common challenges such as overfitting, limited data, and lack of prior work on geometry-aware architectures.	X	X	X

Table 13. Summary of findings

4.7 Context diagram

The context diagram below illustrates the boundaries and interactions between the system and the users.

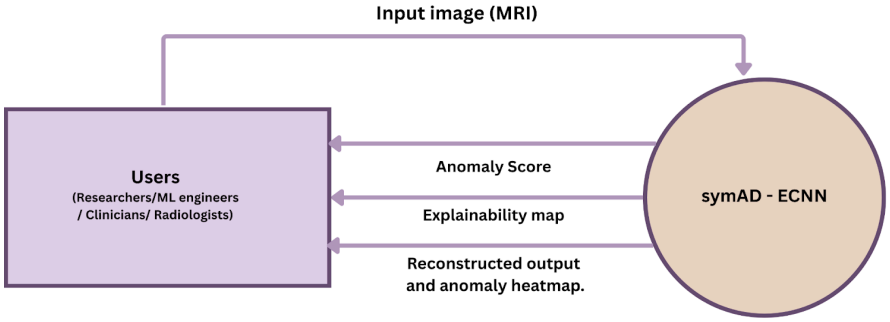


Figure 5. Context diagram (self-composed)

4.8 Use case diagram

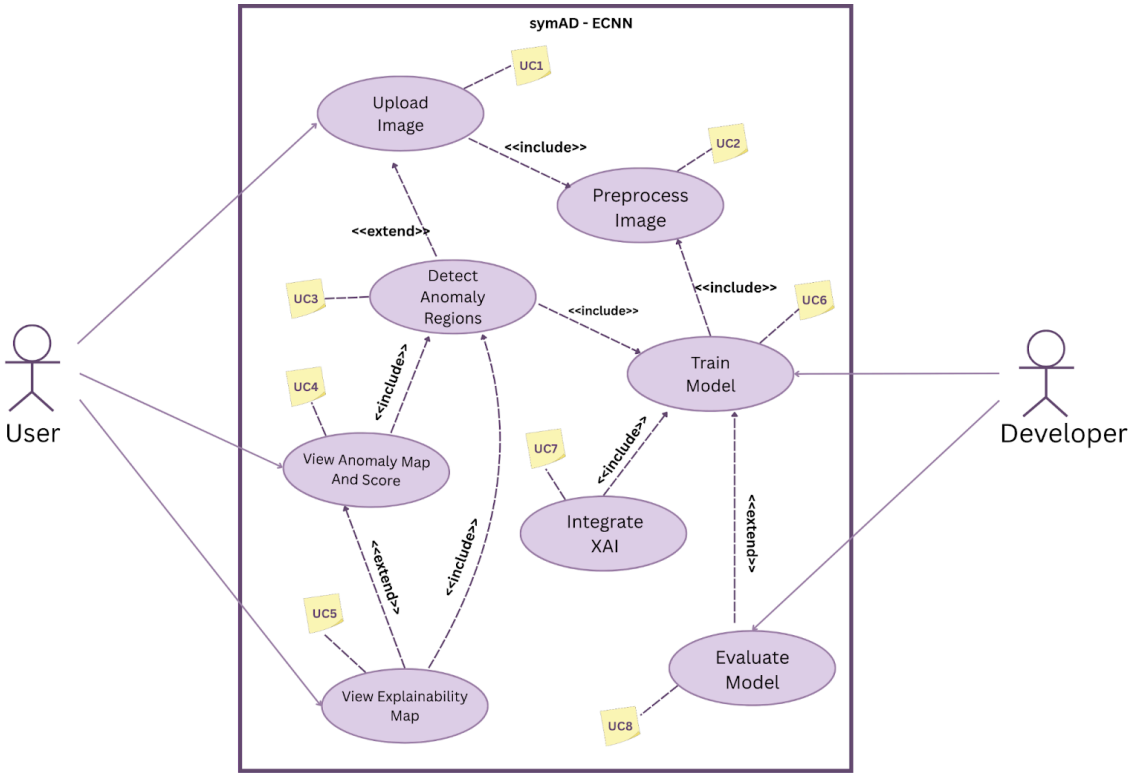


Figure 6. Use case diagram (self-composed)

4.9 Use case descriptions

Specified below is one of the main use case descriptions, while the continuation can be found in [Appendix C.2.](#)

Use Case Name	View Anomaly Map and Score (UC4)
Description	The user views the anomaly map and score output generated by the model.
Participating Actors	User
Pre Conditions	Anomaly detection process must be completed.
Extended Use Cases	View explainability maps (UC5)
Included Use Cases	Detect Anomaly Regions (UC3)
Main Flows	1. The system retrieves the output from the model. 2. The anomaly heatmap and score are generated and displayed. 3. The user views the map, identifies highlighted regions, and interprets anomaly scores.
Alternate Flows	The user may compare the anomaly map with a baseline or reference scan for validation.
Exceptional Flows	No anomaly detected.
Post Conditions	The user can also view explainability maps to interpret better results.

Table 14. Use case description

4.10 Requirements

The 'MoSCoW' principal technique was used to prioritize the functional and non-functional requirements. A description on this can be found in **Appendix C.3.**

4.10.1 Functional requirements

ID	Description	Priority (MoSCoW)
FR1	The system must allow input of MRI images for processing and anomaly detection.	Must have
FR2	Images must be normalized, resized and preprocessed for model input.	Must have
FR3	The system must support training of the proposed architecture on healthy and annotated MRI data.	Must have

FR4	The model must identify anomalous regions in MRI scans and generate corresponding anomaly scores.	Must have
FR5	Incorporate explainable AI techniques such as Grad-CAM, for transparency.	Must have
FR6	Display the anomaly maps and scores for analysis and benchmarking.	Should have
FR7	Visualize the explainability overlays to assess the model's explainability.	Should have
FR8	Quantitatively compare performance against baseline models using defined metrics.	Should have
FR9	Build a minimal prototype interface to demonstrate upload, detection, and visualization capabilities for testing and validation.	Should have
FR10	The model won't be trained and tested on other modalities and organs/domains.	Will not have
FR11	The system won't be deployed for real time clinical usage.	Will not have

Table 15. Functional requirements

4.10.2 Non-functional requirements

ID	Non-Functional Requirement	Description	Priority (MoSCoW)
NFR1	Accuracy	The model must achieve high anomaly detection accuracy (measured via AUROC, Dice, F1-score).	Must have
NFR2	Generalizability	The model should maintain performance across multiple MRI datasets (e.g., BraTS, IXI).	Must have
NFR3	Computational efficiency	Training and inference should be feasible on available GPU resources within reasonable time.	Should have

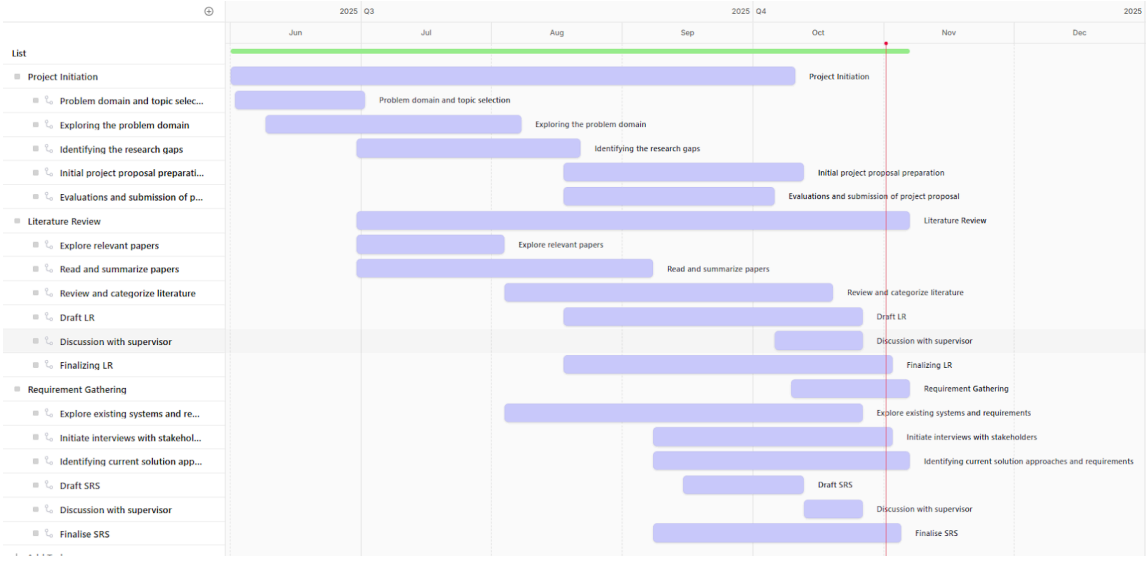
NFR4	Reproducibility	Experiments should be repeatable with documented parameters, dataset splits, and configurations.	Should have
NFR5	Usability	The prototype interface should be simple and intuitive enough for demonstration and researcher evaluation.	Should have
NFR6	Security and privacy	Although no patient data is directly collected, all handling of MRI files could avoid unnecessary storage or exposure.	Could have

Table 16. Non-functional requirements

4.11 Chapter summary

The chapter began with a discussion of the system stakeholders, illustrated through a rich image diagram, an onion model, and descriptions, providing a comprehensive understanding of their roles and relationships, followed by the use case diagram, descriptions and the context diagram. Also, the key findings on the results from the several selected methodologies of requirement elicitation was discussed. The 'MoSCoW' technique was used to break down the functional and non-functional needs at the end of the chapter.

5. Time Schedule



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APPENDIX A - LITERATURE REVIEW

A.1 DL advancements over the years

Time / Focus	Approach / Technique	Description
Pre-2020	**Traditional and CNN-based Models**	Conventional machine learning struggled with high-dimensional imaging data. CNNs became dominant for supervised tasks (e.g., tumour classification) due to their strong spatial feature extraction, but required large labeled datasets.
2020–2022	**Early DL AD (Autoencoders, CNN-AE hybrids)**	Autoencoders introduced unsupervised AD by learning to reconstruct normal brain structures and identifying anomalies through reconstruction error. CNN-AE hybrids improved spatial encoding and feature richness.
2023–2025	**Advanced DL AD (GANs, Diffusion, Hybrid Models)**	Systematic categorization into reconstruction-based (AEs, GANs) and prediction-based (CNN classifiers) models. GANs and diffusion models mitigated data imbalance and small-sample limitations.
2024–2025	**Brain AD Focus (CNN-RNN, CNN-SVM, XAI Integration)**	Hybrid frameworks combining CNNs with RNNs or SVMs enhanced temporal and decision-level robustness. XAI tools are increasingly used to improve interpretability and clinical trust in brain MRI anomaly detection.

Table . Deep learning advancements over the years

A.2 Comparison of Major Architectures for Medical Image Anomaly Detection

Category	Representative Models	Core Architecture	Key Strengths	Limitations
Autoencoders (AE, VAE, GAN-based)	Baur et al. (2020, 2021); Javed et al. (2024); Lu et al. (2023); Luo et al. (2023)	Encoder–decoder (symmetrical) with pixel-wise reconstruction	Simple, unsupervised; captures normality priors	Sensitive to rotation/orientation; overfits on small datasets; limited interpretability
Hybrid CNN–Transformer	Meng et al. (2025) (AFC-Unet); He et al. (2023) (HCTNet); Liu et al. (2024) (STA-Former); Gao et al. (2025) (PFormer)	CNN encoder + Transformer decoder or attention gates	Captures both local and global context; strong segmentation accuracy	High computational cost; data-hungry; prone to overfitting
Self-Supervised and Translation Consistent	SALAD (Baur et al., 2021); Li et al. (2023)	Encoder–decoder with self supervision and translation consistency	Improved robustness; effective for label-scarce data	Limited structural interpretability; complex training setup

Geometry-Aware / Equivariant CNNs	Yan Li et al. (2020) (Dy-GCNNs); Pang et al. (2022) (GER-UNet); Lafarge et al. (2020); Winkels & Cohen (2019)	Group-equivariant convolutional layers	Invariant to rotation/reflection; reduced augmentation need; improved generalization	Computationally expensive; limited adoption in 3D settings
Proposed ECNN-AE	*Current study*	Equivariant CNN-based Autoencoder	Geometry-preserving reconstruction; interpretable latent space; data-efficient	Implementation complexity; requires group selection tuning

Table. Comparison of major architectures

A.3 Concept map

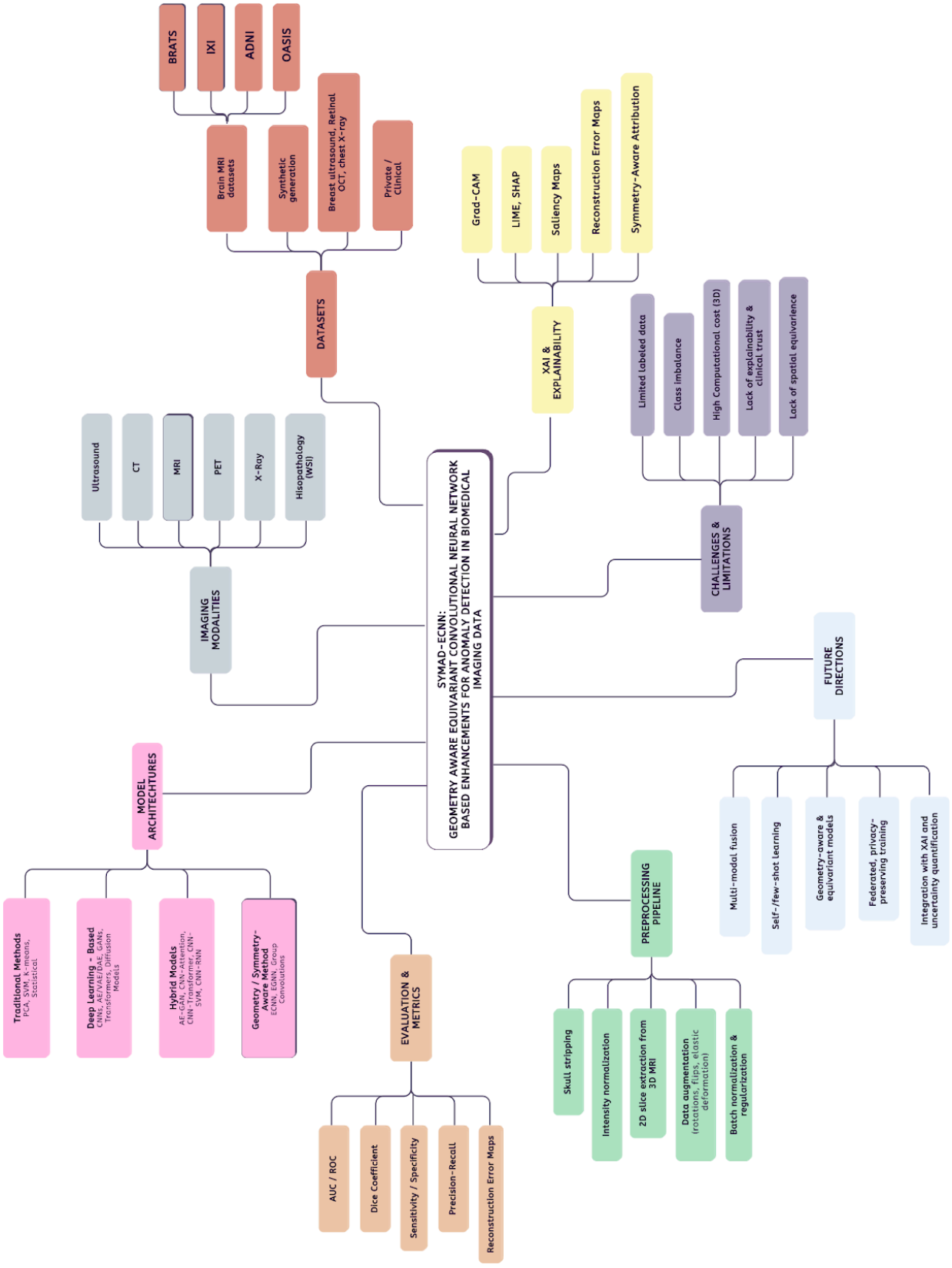

Figure . Concept map(self composed)

APPENDIX B - METHODOLOGY

B.1 Gantt chart

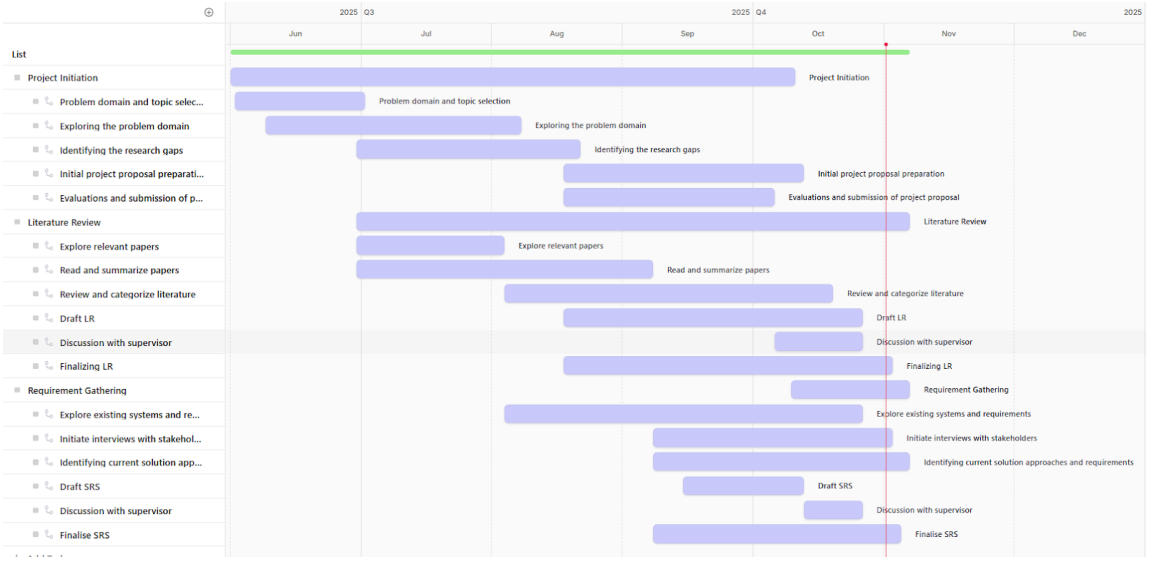

Figure . Gantt chart (self composed)

APPENDIX C - SRS

C.1 Interviewee details

The interviewees along with their details are given below.

Mr. Savin Madapatha	Graduate researcher in deep learning and neurological disorders. (University Of Westminster)
Mr. Nuwin Godakanda Arachchi	Research supervisor in areas of deep learning and medical imaging. ((University Of Westminster))
Mr. Tirush Dumil	Graduate researcher in deep learning and biomedical engineering. (University of Sri Jayewardenepura)
Ms. Ramisa Farha	Graduate researcher in deep learning and neuro medical imaging. (Morgan State University)

Table . Interviewee details

C.2 Use case descriptions

Use Case Name	**Upload Image (UC1)**
Description	The user uploads an image (brain MRI) for anomaly detection.
Participating Actors	User
Pre Conditions	Users must be authenticated and the system interface should be active.
Extended Use Cases	Detect Anomaly Regions (UC3)
Included Use Cases	Preprocess Image (UC2)
Main Flows	1. User uploads an image (brain MRI) to the system 2. The system validates image initiates preprocessing
Alternate Flows	None
Exceptional Flows	If the image format is unsupported, the system displays an error and prompts for a new file to be uploaded.
Post Conditions	The image is successfully uploaded, validated and preprocessed for anomaly detection.

Use Case Name	Train Model (UC6)
Description	The developer trains the symAD-ECNN model on biomedical datasets (IXI) to learn normal anatomical features, so that it can be used to detect and visualise anomalies.
Participating Actors	Developer
Pre Conditions	Data must be preprocessed as per the input requirements.
Extended Use Cases	Evaluate Model (UC8)
Included Use Cases	Preprocess Image (UC2)
Main Flows	1. The developer initiates training using the preprocessed dataset. 2. The model encodes inputs using group-equivariant convolutions to capture spatial symmetries. 3. The latent representation is decoded to reconstruct the original input. 4. The reconstruction error is computed for each pixel/voxel. 5. The model parameters are updated iteratively to minimize reconstruction loss. 6. The trained model is stored for further anomaly detection tasks.
Alternate Flows	If libraries are incompatible, equivalent group-equivariant implementations are substituted to ensure functionality.
Exceptional Flows	Resume or restart training, if interrupted due to resource failure.
Post Conditions	The model is ready to detect anomalous regions and evaluations.

Use Case Name	Detect Anomaly Regions (UC3)
Description	The system runs the model to identify potential anomalies within the (MRI) image.
Participating Actors	System
Pre Conditions	Image must be loaded and preprocessed
Extended Use Cases	None
Included Use Cases	Train Model (UC6)
Main Flows	1. The system loads the trained model. 2. Processes the preprocessed image. 3. Computes anomaly scores or reconstruction errors. 4. Generates a visual anomaly heatmap highlighting regions of deviation.
Alternate Flows	Fall back into baseline models.

Exceptional Flows	Display model error, if the model is not found or corrupted.
Post Conditions	Anomaly map, explainability overlays, and quantitative anomaly scores are generated successfully.

Table . Use case descriptions (continued)

C.3 MoSCoW' Priority Levels

Priority Level	Description
M - Must have	These are the core elements that serve as the project's foundation and are required for its effective completion.
S - Should have	These features are important but not mandatory for the prototype's core functionality. Yet, it would greatly enhance the project's value.
C - Could have	These are the features that are desirable but not necessary for the core functionality. These can be incorporated if time persists.
W - Will not have	These requirements are the least crucial and are not in-scope. They might be taken into account for future enhancements.

Table. MoSCoW' priority levels